

ECO4102Z test solutions (2024)

Question One (MCQ)

1. A
2. C - A high RRA coefficient implies that the investor exhibits decreasing relative risk aversion, meaning they will allocate less to risky assets as their wealth increases.
3. C
4. B
5. D - From CAPM, the fair expected return = $8 + 1.25 \times (15 - 8) = 16.75\%$
Actual return = 17%
 $\alpha = 17 - 16.75 = 0.25\%$

Question 2 (stock indices)

2.1. $r_A = (532.2 + 0.63 - 504) / 504 * 100 = 5.72\%$ (0.5)

$$r_B = (51.50 + 0.80 - 42) / 42 * 100 = 24.52\% \text{ (0.5)}$$

$$r_C = (116.1 + 1.1 - 126) / 126 * 100 = -6.98\% \text{ (0.5)}$$

$$r^e = (5.72 + 24.52 - 6.98) / 3 = 7.75\% \text{ (0.5)}$$

[2]

2.2.

$$Index^E = \frac{\sum_{i=1}^N Q_i^* P_i}{d}$$

Lowest common factor = 504

$$504 * 1 + 42 * 12 + 126 * 4 = 1512$$

$$Index = 1512 / 16 * 1.0775 = \mathbf{101.82}$$

Or

$$(532.2 + 0.63) * 1 + (51.50 + 0.8) * 12 + (116.1 + 1.1) * 4 = 1629.231$$

$$r^e = 1629.23 / 16 = \mathbf{101.83}$$

[4]

2.3. This statement is true. (1) Tracking an equal-weighted index involves high transaction costs because the turnover of stocks is high (buying and selling) to ensure that the stocks are equal-weighted. (1)

2.4. Weighting of assets in mkt cap index = 63.12% (A), 15.6% (B), 21.28% (C) (1)

$$Return = (0.6312 \times 5.72) + (0.1560 \times 24.52) + (0.2128 \times -6.98) = 5.95\% \text{ (1)}$$

2.5. Index rebalancing would occur when market prices change, the the weightings of the shares differs from an equal weighting – therefore the fund manager will need to purchase and sell shares so that the index goes back to its equal weighting. (1) (1)

Index reconstitution occurs when the constituent shares in an index are changed (either removed, or added). Therefore, the fund manager will need to sell all the shares of a company which is removed from the index, and buy proportionate shares of the company which is getting included.

(1) (1)

[4]

Question Three

1. The first derivative of John's utility function $u(I)=I^2$ is equal to $u'(I)= 2I$ and the second derivative $u''(I) = 2$, which is positive. Therefore a positive second derivative indicates that the utility function is convex. This implies risk loving behaviour, which means that he finds greater joy in potential high payoffs, even if they are uncertain.
2. John's utility with an income of R30 000 = $u(30) = (30)^2 = 900$
 Expected utility = $[0.4 \times u(50)] + [0.6 \times u(10)] = [0.4 \times 2500] + 0.6 \times 100 = 1060$
 John's current utility is 900, and the expected utility of the business opportunity is 1060. Therefore since this value is greater, John should take on the business opportunity.
3. Increasing relative risk aversion occurs when an investor becomes more wealthy, and decreases the percentage of wealth that she places in risky assets (ie. Becomes more risk averse).

Question Four

- 4.1. Not possible. (1) The reward-to-variability ratio for Portfolio A is better than that of the market (1) (1). This scenario is impossible according to the CAPM because the CAPM predicts that the market is the most efficient portfolio (1). Using the numbers supplied:

$$S_A = \frac{.16 - .10}{.12} = 0.5 \quad S_M = \frac{.18 - .10}{.24} = 0.33$$

Portfolio A provides a better risk–reward trade-off than the market portfolio.

4.2.

- a. Stock A has higher firm-specific risk because the deviations of the observations from the SCL are larger for Stock A than for Stock B. Deviations are measured by the vertical distance of each observation from the SCL.
- a. Beta is the slope of the SCL, which is the measure of systematic risk. The SCL for Stock B is steeper; hence Stock B's systematic risk is greater.
- b. Alpha is the intercept of the SCL with the expected return axis. Stock A has a small positive alpha, whereas Stock B has a negative alpha; hence, Stock A's alpha is larger.

4.3.

$$w_1 = \frac{R_1\sigma_2^2 - R_2\sigma_{1,2}}{R_1\sigma_2^2 + R_2\sigma_1^2 - (R_1 + R_2)\sigma_{1,2}}$$

$$cov = -0.1 \times \sqrt{0.25} \times \sqrt{0.5} = -0.0354(1)$$

$$W_x = \frac{(0.08)(0.5) - (0.11)(-0.0354)}{(0.08)(0.5) + (0.11)(0.25) - [(0.08 + 0.11) \times -0.0354]}(1m) = \frac{0.0439}{0.0742} = 59.16\%(1)$$

$$\text{Therefore } W_y = 40.84\% (1)$$

$$E(R) = (0.5916 \times 0.12) + (0.4084 \times 0.15)(0.5) = 13.23\% (1)$$

σ_p

$$= \sqrt{(0.5916^2 \times 0.25)(0.5) + (0.4084^2 \times 0.5)(0.5) + (2 \times 0.5916 \times 0.4084 \times 0.25 \times 0.71 \times -0.1)(0.5)}$$

$$= \sqrt{0.1623} = 40.29\%(1)$$

4.4.

- a. The intercept is statistically insignificant (1), beta is statistically significant (1) and positive (implying positive relationship between risk and return) (1), R square is low at 39% however which indicates that whilst beta has explanatory power, there may be other factors which are relevant.(1), Therefore there is mixed/weak support for the CAPM model(1).
- b. I disagree – the beta of a hedge is less than 1. Beta of sasol is greater than 1, which indicates it is an aggressive stock. (1). (1).

Question Five

5.1. C (1)- Jordan Lee's statement is **incorrect** because recommending a standard, one-size-fits-all investment policy based solely on categories or age groups does not account for Alex's unique circumstances, financial goals, risk tolerance, and personal preferences. Investment Policy Statements (IPS) are most effective when they are customized to an individual's specific needs and objectives rather than relying on general templates. (1)

Patricia Roy's statement is **correct** because it highlights the importance of flexibility in investment policy. Markets and personal circumstances can change over time, so a rigid policy might limit the advisor's ability to adapt to new conditions. However, while flexibility is important, it's also crucial to have a structured IPS that provides a clear strategy while allowing adjustments as needed. (1)

5.2. Alex's return objectives are primarily driven by his goal to retire comfortably at age 60(1) and live off his portfolio by withdrawing 4% annually(1). Therefore his portfolio needs to grow enough to provide this 4% annual withdrawal (plus inflation) (1). Therefore assuming a 4% inflation, his portfolio needs to provide a real return of 8% annually to meet his goal on retirement.

5.3. Ability to take on risk - Alex's ability to take on risk is relatively high, given his current financial situation. He has a substantial portfolio worth \$1.8 million, no immediate liquidity needs, and a separate emergency fund in cash(1). His business income more than covers his living expenses, reducing the pressure on his investments to generate cash flow in the short term. Additionally, with a 12-year time horizon until retirement, Alex has a moderate-to-long timeframe to recover from potential market downturns, which generally supports a higher capacity for risk. (1)

Alex's willingness to take on risk is more moderate(1). He describes his risk tolerance as "average," which suggests a preference for a balanced approach rather than extreme risk. His surprise at learning about the volatility of small-cap stocks also indicates a possible gap between his perceived risk tolerance and the actual risks in his current portfolio. Although he's willing to accept some fluctuations for higher returns, he may not be comfortable with high levels of volatility or significant losses, which should be factored into his IPS to ensure alignment with his comfort level. (1)

5.4. Alex does not face **significant liquidity constraints**, because: (1)

- Alex has stated that his income from his architecture business is more than sufficient to cover his current living expenses. This means he does not need to rely on his investment portfolio to meet day-to-day expenses.
- Alex has a dedicated emergency fund in cash, which provides a financial cushion for unexpected expenses. This fund reduces the need for liquid assets in his main investment portfolio, allowing him to invest in assets with lower liquidity if desired. (1)

Alex does have a **time constraint**(1) related to his retirement goal, because he plans to retire at age 60, which is 12 years from now. This medium-term horizon provides enough time to take some risk in his portfolio for growth, but it also requires that he be cautious as he approaches retirement to ensure his portfolio is not overly exposed to market volatility. (1)

5.4. Any 2 of the below, one mark each: (students could have also answered with recommendations specific to Alex's situation)

- Purpose or philosophy of the IPS
- Asset allocation
- Actuarial assumptions
- Strategy for meeting investment objectives
- Permissible and restricted investments.